

Program 8: Maven Web App

Step 1: Create the project using the command :

“mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenWebApp01 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false”

```
rakshith@rakshith-VirtualBox:~/Devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenWebApp01 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.0/maven-archetype-webapp-1.0.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.0/maven-archetype-webapp-1.0.pom
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/rakshith/Devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenWebApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/rakshith/Devops/MyMavenWebApp
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 7.073 s
[INFO] Finished at: 2026-04-08T08:35:11+05:30
```

Step 2: Using tree command to view the structure

```
rakshith@rakshith-VirtualBox:~/Devops$ tree
.
├── pom.xml
└── src
    ├── main
    │   ├── resources
    │   └── webapp
    │       ├── index.jsp
    │       └── WEB-INF
    │           └── web.xml
└── 5 directories, 3 files
```

STEP 3: Update the pom.xml by adding the following dependencies and plugins

```
<!-- Servlet API for Java EE -->
    <dependency>
        <groupId>javax.servlet</groupId>
        <artifactId>javax.servlet-api</artifactId>
        <version>4.0.1</version>
        <scope>provided</scope>
    </dependency>

<build>
    <plugins>
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-war-plugin</artifactId>
            <version>3.4.0</version> <!-- Update to latest stable version -->
        </plugin>
    </plugins>
    <finalName>MyMavenWebApp</finalName>
</build>
```

STEP 4: run mvn clean install

```

[INFO] Stopped
[INFO] Deleted pom.xml
rakshith@rakshith-VirtualBox:~/Devops/MyMavenWebApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MymavenWebApp01 >-----
[INFO] Building MymavenWebApp01 Maven Webapp 1.0-SNAPSHOT
[INFO] -----[ war ]-----
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/r
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/r
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/r
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/r
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/r
Downloading from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servl
Downloaded from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servle
Downloading from central: https://repo.maven.apache.org/maven2/net/java/jvnet-parent/3/
Downloaded from central: https://repo.maven.apache.org/maven2/net/java/jvnet-parent/3/jv
Downloading from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servl
Downloaded from central: https://repo.maven.apache.org/maven2/javax/servlet/javax.servle
[INFO]

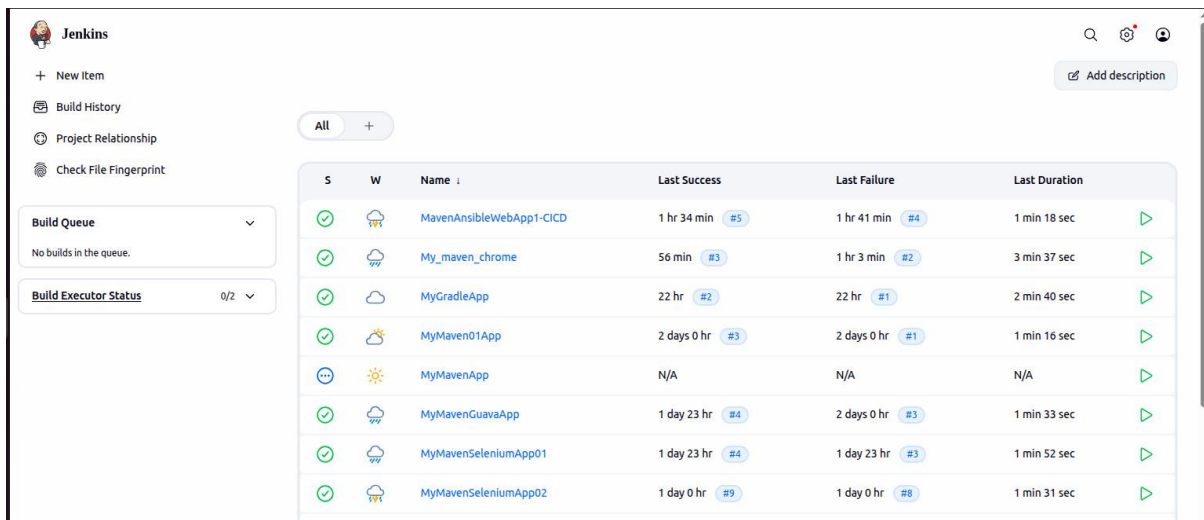
```

STEP 5: Verify tree structure

```
rakshith@rakshith-VirtualBox:~/Devops$ tree
.
├── pom.xml
├── src
│   ├── main
│   │   ├── resources
│   │   └── webapp
│   │       ├── index.jsp
│   │       └── WEB-INF
│   │           └── web.xml
└── target
    ├── classes
    ├── maven-archiver
    │   └── pom.properties
    ├── MymavenWebApp01
    │   ├── index.jsp
    │   ├── META-INF
    │   ├── WEB-INF
    │   │   ├── classes
    │   │   └── web.xml
    └── MymavenWebApp01.war
```

Jenkins Pipeline for MavenWebApp:

Step 1: Create a New Pipeline Job



The screenshot shows the Jenkins dashboard with a sidebar on the left containing links for 'New Item', 'Build History', 'Project Relationship', and 'Check File Fingerprint'. The main area displays a table of jobs. The table has columns for 'S' (Status), 'W' (Icon), 'Name', 'Last Success', 'Last Failure', and 'Last Duration'. The jobs listed are:

S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	MavenAnsibleWebApp1-CICD	1 hr 34 min #5	1 hr 41 min #4	1 min 18 sec
✓	☁	My_maven_chrome	56 min #3	1 hr 3 min #2	3 min 37 sec
✓	☁	MyGradleApp	22 hr #2	22 hr #1	2 min 40 sec
✓	☁	MyMaven01App	2 days 0 hr #3	2 days 0 hr #1	1 min 16 sec
⚙	☀	MyMavenApp	N/A	N/A	N/A
✓	☁	MyMavenGuavaApp	1 day 23 hr #4	2 days 0 hr #3	1 min 33 sec
✓	☁	MyMavenSeleniumApp01	1 day 23 hr #4	1 day 23 hr #3	1 min 52 sec
✓	☁	MyMavenSeleniumApp02	1 day 0 hr #9	1 day 0 hr #8	1 min 31 sec

Go to Dashboard → + New Item.

Enter name, select Pipeline, and click OK.

New Item

Enter an item name

MyMavenWebApp001

Select an item type



The 'New Item' dialog shows five options for item types:

- Pipeline**: Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**: Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Maven project**: Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
- Multi-configuration project**: Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**

An 'OK' button is at the bottom.

Step 2: Configure Pipeline from SCM

Scroll down and select Pipeline script from SCM.

SCM: Git Enter repository URL and select GitHub credentials.

Jenkins / All / MymavenWebApp01 / Configure

Configure

- General
- Triggers
- Pipeline**
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

https://github.com/rakshith31-hub/MyMavenWebApp01.git

Credentials

GitHub Credential + Add

Advanced

Save Apply

Step 3: Create Jenkinsfile

```
pipeline {
    agent any // Use any available agent

    tools {
        maven 'Maven' // Ensure this matches the name configured in
        Jenkins
    }

    stages {
        stage('Check
        out') {
            steps {
                git branch: 'main', url:
'https://github.com/rakshith31-
                hub/MyMavenWebApp.git'
            }
        }

        stage('Build')
        { steps {
            sh 'mvn clean package' // Run Maven build
        }
        }

        stage('Test')
        { steps {
            sh 'mvn test' // Run unit tests
        }
        }
    }
}
```

```

stage('Deploy WAR')
{
    steps {
        sh 'cp target/MymavenWebApp01.war
/opt/tomcat/webapps/'
    }
}

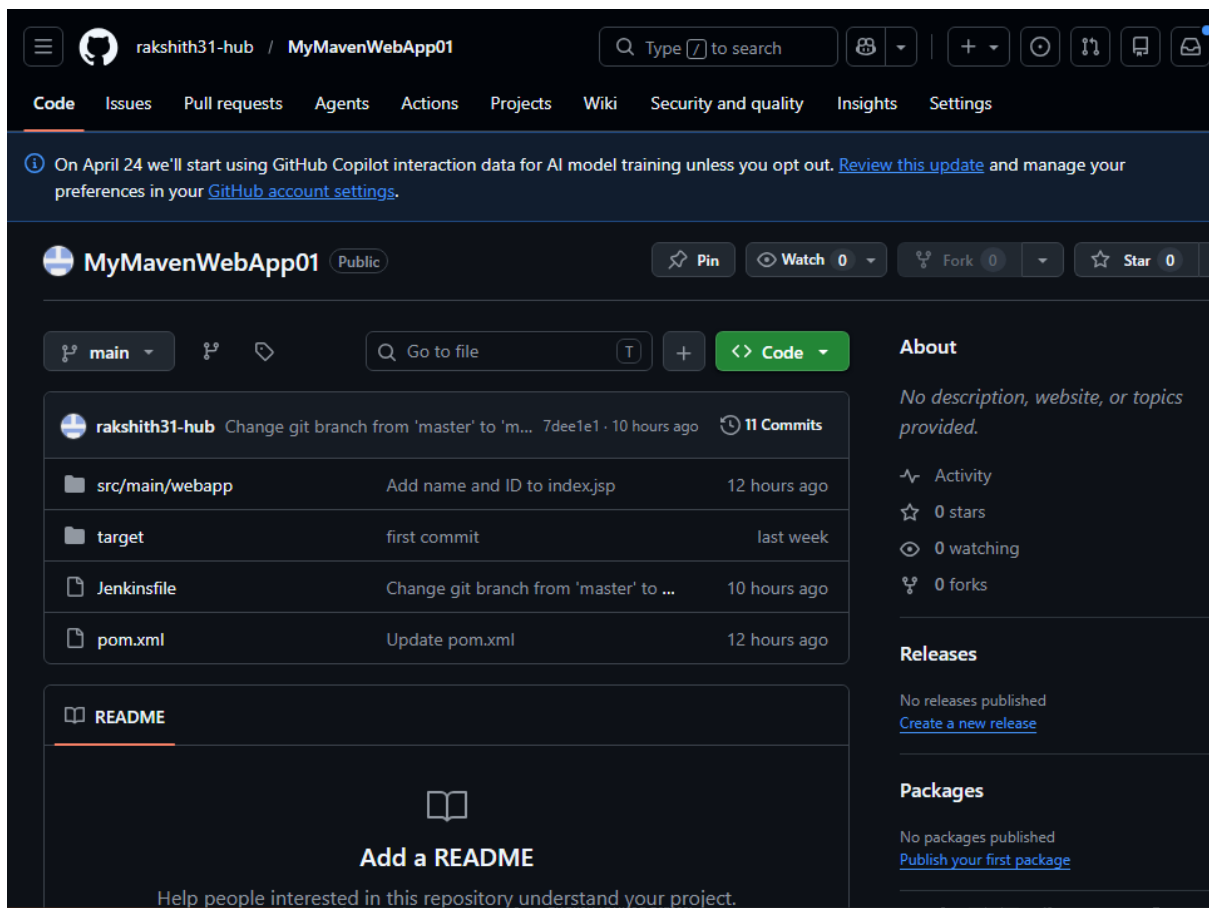
}

post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }
}

}

```

Step 4: Now Push changes and commit by pushing the necessary “Jenkinsfile” to the respective Repo



Step 5: Build the Pipeline

Go to Jenkins and click Build Now.

Monitor Build Stages Use Stage View and Full Stage View to monitor execution

